

Analyzing Global COVID-19 Vaccination Disparities Using a Spark-Based Data Lakehouse Pipeline

Integrating Multi-Source Health, Demographic, and Policy Data with K-Means Clustering

Nurul Adriana Binti Kamal Jefri
Faculty of Computing
Universiti Teknologi Malaysia (UTM)
Johor Bahru, Johor, Malaysia
adriana04@graduate.utm.my

Abstract—The global rollout revealed stark inequities in COVID-19 vaccination rates among high-income and low-income countries, but rates are not typically associated with the size of the population or disease burden, or with government policy actions. In this study, a data pipeline is proposed that combines COVID-19 vaccination data, COVID-19 cases and deaths data, population data, and COVID-19 Government Response Tracker (OxCGRT) stringency index into a single table that is stored in a data lakehouse using bronze-silver-gold medallion architecture and ready for analysis. Once the four sources are cleaned and merged on country code and date, a machine learning (ML) model created using K-Means clustering from Spark MLlib clusters countries into one of the three tiers of vaccination progress based on vaccination rate per capita, country trend of case fatality, and policy stringency. The resulting data set and cluster assignments are meant to inform an interactive dashboard to identify countries that are implementing vaccines slowly and the context of those delays.

Keywords—component; *COVID-19 vaccination; data lakehouse; medallion architecture; K-Means clustering; data integration; global health disparity; Apache Spark.*

COVID-19 triggered the biggest vaccination campaign ever, which involved governments procuring and administering vaccines at an unprecedented speed. Even with this coordinated effort, however, there was widespread variation in coverage across the countries – in the first two years of the campaign, more than 70% of the population in high income countries had been fully vaccinated, while approximately 30% had been fully vaccinated in low income countries (Ferranna, 2023). This gap is not just one of logistics, but also due to supply constraints, distribution networks, the quality of local health systems, and vaccine hesitancy (Kunyeje et al., 2023). The numbers of raw vaccinations are usually provided by country and by date and disseminated on a public dashboard that monitors vaccination progress, such as the dataset released by Our World in Data and replicated on Kaggle. Useful as they are, they are seldom associated with the context required to understand them, such as population size to calculate per-capita rates, case and death counts to gauge the severity of the pandemic, or the stringency of government policy to grasp the regulatory context in which the rollout is taking place. Given this, it is hard to judge whether a country's vaccination pace was due to limited resources, lower disease burden, or less stringent containment policy or any mixture of

I. INTRODUCTION

these three. This study is filling that gap by having a Spark-based data pipeline bring together four public datasets and put them into a single store in which they are ready for analysis; and by using an unsupervised learning algorithm to classify countries by their vaccination performance. The specific objectives are;

(i) A data pipeline is designed and implemented to extract, clean and integrate COVID-19 vaccination, COVID-19 case and death, population and COVID-19 government policy stringency data from various sources using Apache Spark.

(ii) It is suggested to use a data lakehouse organization, with bronze, silver, and gold layers, to store a variety of both raw and curated versions of the integrated data.

(iii) Spark MLlib is then used to cluster countries according to their uptake, fatality rates, and policy stringency to create vaccination-progress tiers

The rest of this paper is organized as follows. The current state of the art for global vaccine inequity and for data pipeline and lakehouse architectures for health-related data is summarized in Section II. The method described, data sources, data preparation steps as well as the system architecture are described in Section III. The planned implementation is described in Section IV and the paper is finished with a summary of expected contributions in Section V.

II. Literature Review

A. Global COVID-19 Vaccine Inequity

Several studies have documented and explained the unequal distribution of the COVID-19 vaccines around the world. In low-income countries, particularly in Africa, disparities in vaccine coverage were examined by Kunyenje et al. (2023) and they found that the major factors responsible for disparities were inadequate supply, unequal distribution, poor local manufacturing capabilities, poor health system infrastructure, and vaccine hesitancy. It was also demonstrated through their analysis that there are differences between high and low income countries, but there may also be differences within low income countries, with rural populations often being excluded when compared

with urban populations. Ferranna (2023) further expanded on this analysis by quantifying the health, social, and economic impacts of vaccine inequity, which was driven by vaccine nationalism, IP barriers, limited manufacturing capacity, and healthcare system shortcomings in lower-income areas. In contrast, neither study presents vaccination data with government policy response or case-severity indicators in a structured, queryable format, a gap filled by this study, which presents country-level aggregate data from both studies, mostly from Our World in Data.

B. Data Pipeline and Lakehouse Architectures

From the systems perspective, the data lakehouse is proposed by Zaharia et al. (2021) as a pattern that melds the ease of storing data on a data lake with the management capabilities and query performance of the data warehouses, including support for machine learning workloads and ACID transactions. The traditional two-tier model involves retaining two separate data lakes and data warehouses with repeated ETL jobs, which are issues solved by the single data lakehouse tier, and is argued to add redundant storage costs and data staleness. Additionally, OxCGRT, a continuously updated dataset developed by Hale et al. (2021), was introduced, and in this dataset scores are assigned to each of 19 government actions for over 180 countries. Built mainly for policy research, OxCGRT offers a structured machine-readable indicator of government response which can be linked to epidemiological and demographic information to examine the relationship between policy stringency and public health outcomes.

The literature confirms that there are disparities in vaccination rates around the world and the tools to conduct a study, which require heterogeneous data sources, are available but have not been combined. In none of the existing public resources is the vaccination, case and mortality data combined in a single analytical pipeline, nor is there an automated method to classify countries by their vaccination performance. That gap is filled by this study.

III. METHODOLOGY

A. Data Sources

This study uses four publicly available datasets, summarized in Table I, all of which are country-level and can be joined by ISO-3 country code; three of which also have a common date field, allowing temporal alignment between vaccination progress and case trends and policy stringency.

B. Data Preparation

Each raw data set is first read into Spark as a DataFrame and cleaned separately, by defining column names and removing non-country aggregates (such as continental or income-group totals found in some data sources). No information is dropped, instead only missing information is documented, as missing information provides information about the country's information capacity. Following independent cleaning, the vaccination data, case and death data, and OxCGRt data are linked based on country code only, to calculate vaccination rates and case rates per capita.

C. Data Storage Architecture

The pipeline is followed by a bronze-silver-gold medallion architecture, a pattern that is implemented on a data lakehouse, which brings flexibility and low costs to data lakes, with structure and query performance of a data warehouse (Zaharia et al., 2021). The bronze layer keeps the original (unmodified) form of each of these four datasets intact, including the complete lineage, back to the source. Each of the datasets is transformed as described above to be cleaned and standardized, and stored in the silver layer. Data from the vaccination, case, population, and policy layers are fully integrated with each other on a country-date level and stored in the gold layer, along with the cluster labels generated by the analytical step described below. It's a layered model as opposed to just one relational data warehouse because the underlying sources have different update frequencies, different structures, and aren't always complete; the raw data can be kept in the lakehouse for reprocessing and a clean,

query-ready table can still be exposed for downstream dashboards. The proposed system architecture diagram is presented in Figure 1.

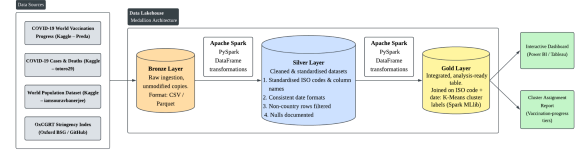


Figure 1. Proposed data lakehouse architecture for the COVID-19 vaccination disparity pipeline.

D. Analytical Technique

A K-Means clustering model, running on Spark MLlib, is added to the gold-layer table to cluster countries in tiers of 'vaccination progress'. The candidate feature set includes per-capita vaccination rate, case-fatality rate and government response stringency over the observation period on average. The analysis does not rank countries individually but groups of countries with similar vaccination, disease-burden and policy profiles are clustered together, highlighting the analysis and making it possible to compare countries across and between income groups.

IV. IMPLEMENTATION

To be completed in the next phase of this project.

A. Authors and Affiliations

To be completed in the next phase of this project.

B. Identify the Headings

To be completed in the next phase of this project.

C. Figures and Tables

To be completed in the next phase of this project.

TABLE I. Table Type Styles

Table Head	Table Column Head		
	Table column subhead	Subhead	Subhead

copy	More table copy ^a		
------	------------------------------	--	--

Table footnote. (Table footnote)

Figure 1. Example of a figure caption.

V. CONCLUSION

To be completed in the next phase of this project.

ACKNOWLEDGMENT

The author would like to thank Dr. Aryati Bakri for guidance and feedback throughout the development of this study.

APPENDIX

To be completed in the next phase of this project.

REFERENCES

- [1] C. A. Kunyenje et al., “COVID-19 vaccine inequity in African low-income countries,” *Frontiers in Public Health*, vol. 11, art. 1087662, 2023.
- [2] M. Ferranna, “Causes and costs of global COVID-19 vaccine inequity,” *Seminars in Immunopathology*, vol. 45, pp. 469–480, 2023.
- [3] T. Hale et al., “A global panel database of pandemic policies (Oxford COVID-19 Government Response Tracker),” *Nature Human Behaviour*, vol. 5, no. 4, pp. 529–538, 2021.
- [4] M. Zaharia, A. Ghodsi, R. Xin, and M. Armbrust, “Lakehouse: A new generation of open platforms that unify data warehousing and advanced analytics,” in *Proc. 11th Conf. Innovative Data Systems Research (CIDR)*, Online, 2021.